

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in global energy and electricity

Source: Our World in Data – Renewable Energy

Data: Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

- Since the Industrial Revolution, fossil fuels have dominated global energy systems.
- Three-quarters of global greenhouse gas emissions come from burning fossil fuels for energy.
- Fossil fuel combustion causes at least 5 million premature deaths per year from local air pollution.
- To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon sources — nuclear and renewable technologies.
- Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources.

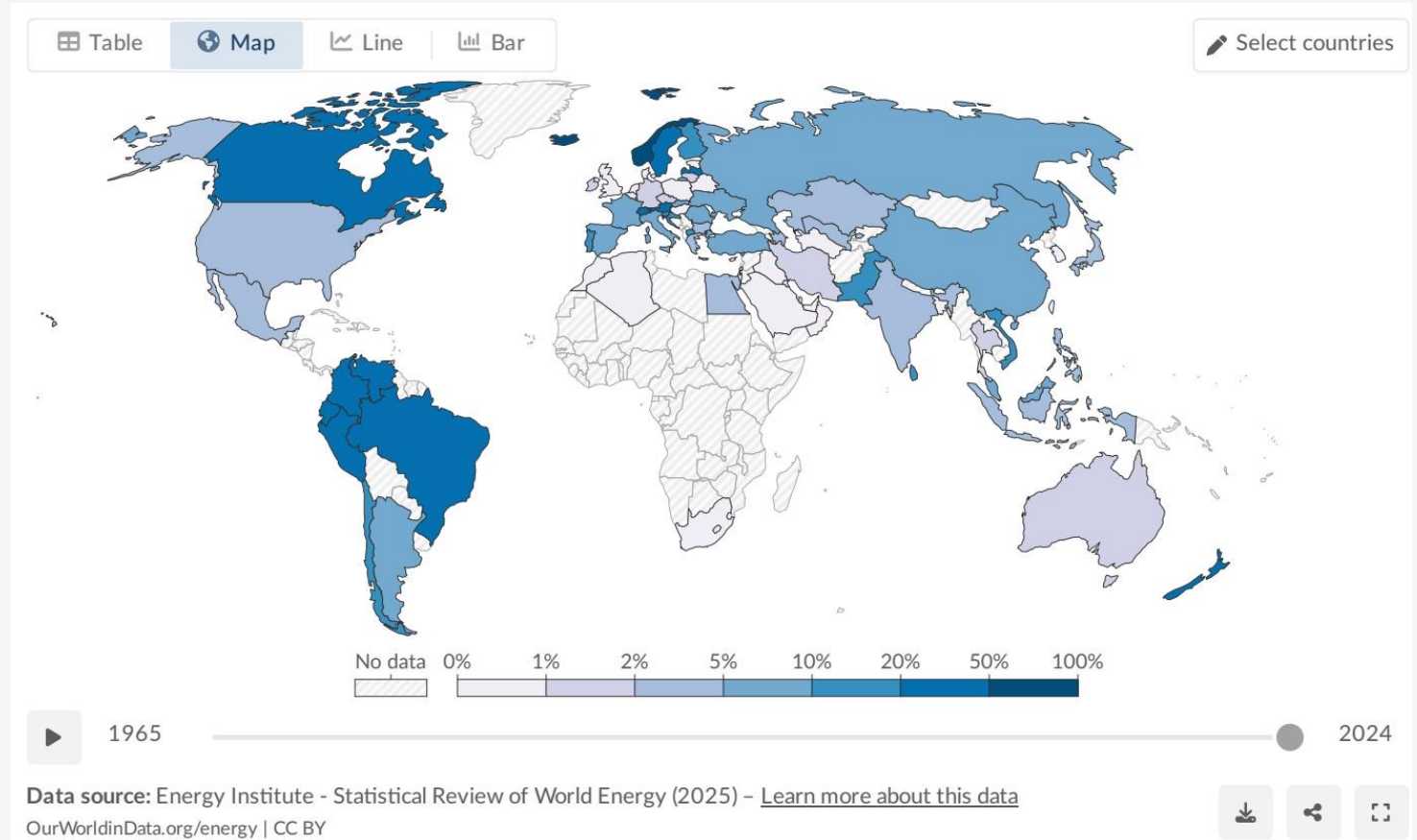
Renewables Now Supply ~14% of Global Primary Energy

Primary energy encompasses electricity, transport, and heating combined (2024 data, substitution method)

Key Insight

Approximately one-seventh of the world's primary energy now comes from renewable technologies.

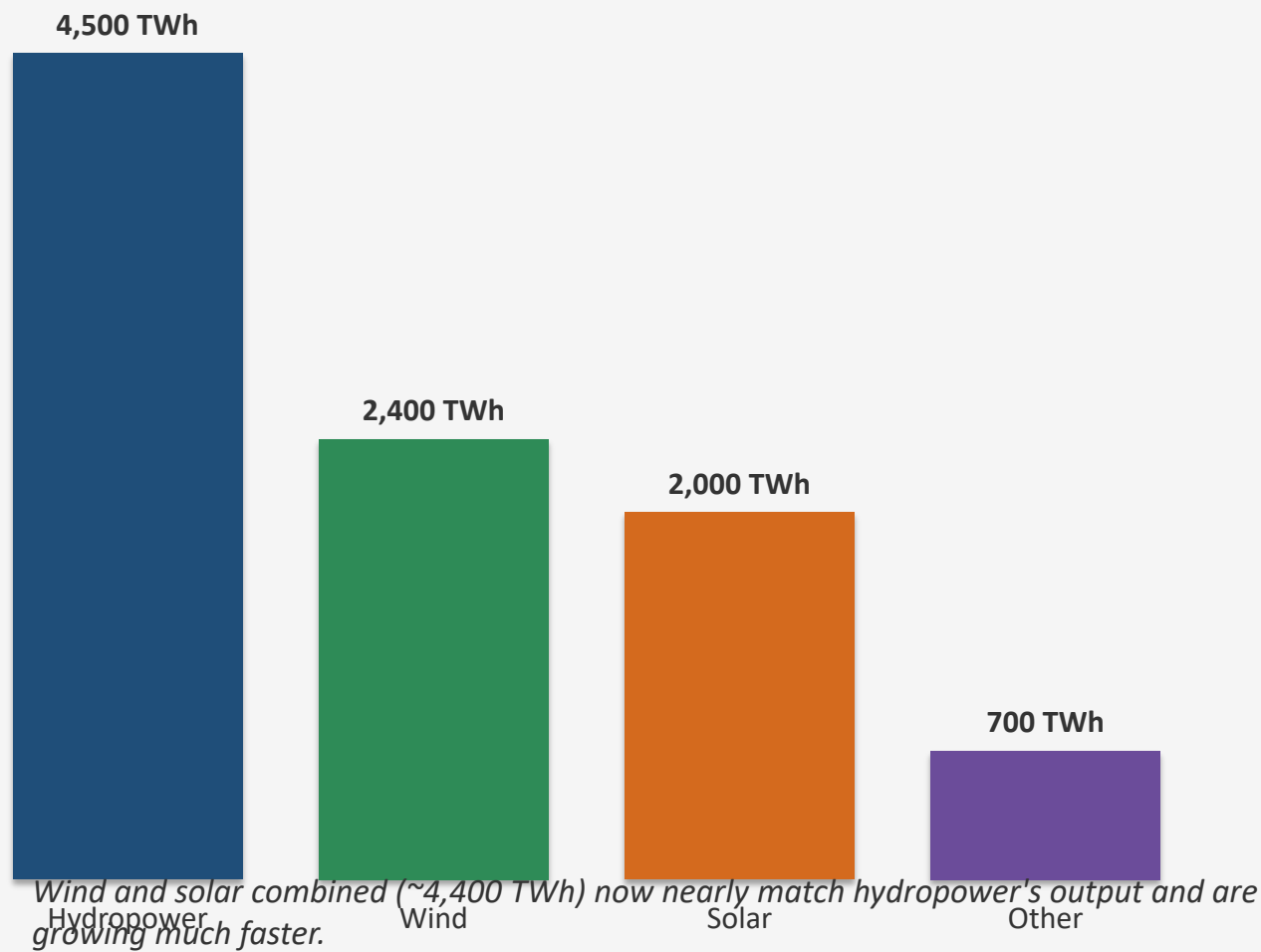
Because transport and heating are harder to decarbonize, the renewable share in total energy is lower than in electricity alone.



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global Renewable Generation by Technology (2024)

Technology	Generation (TWh)	Share of Renewables
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other renewables	~700	~7%



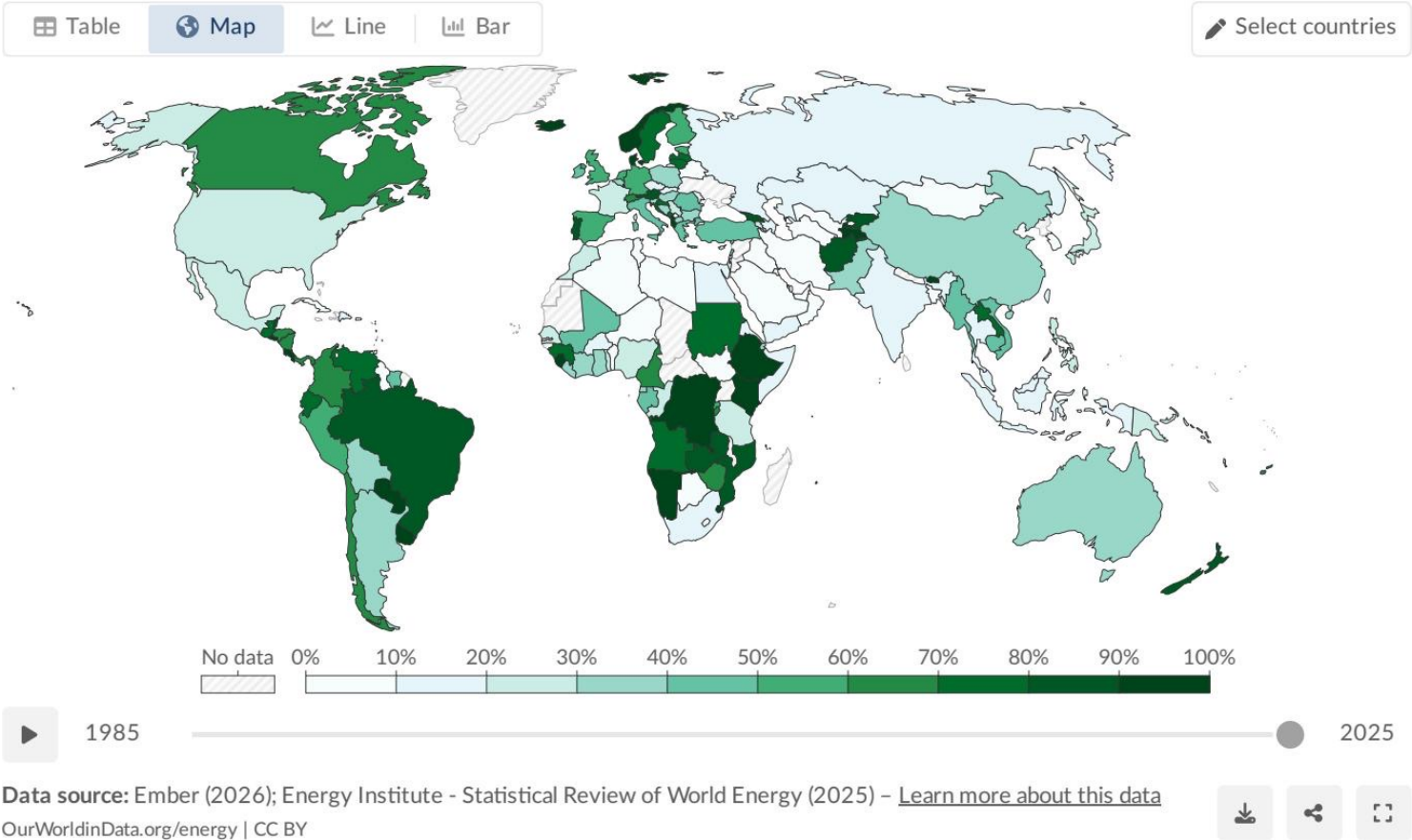
Source: Our World in Data – Renewable Energy (Ember 2026; Energy Institute 2025)

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

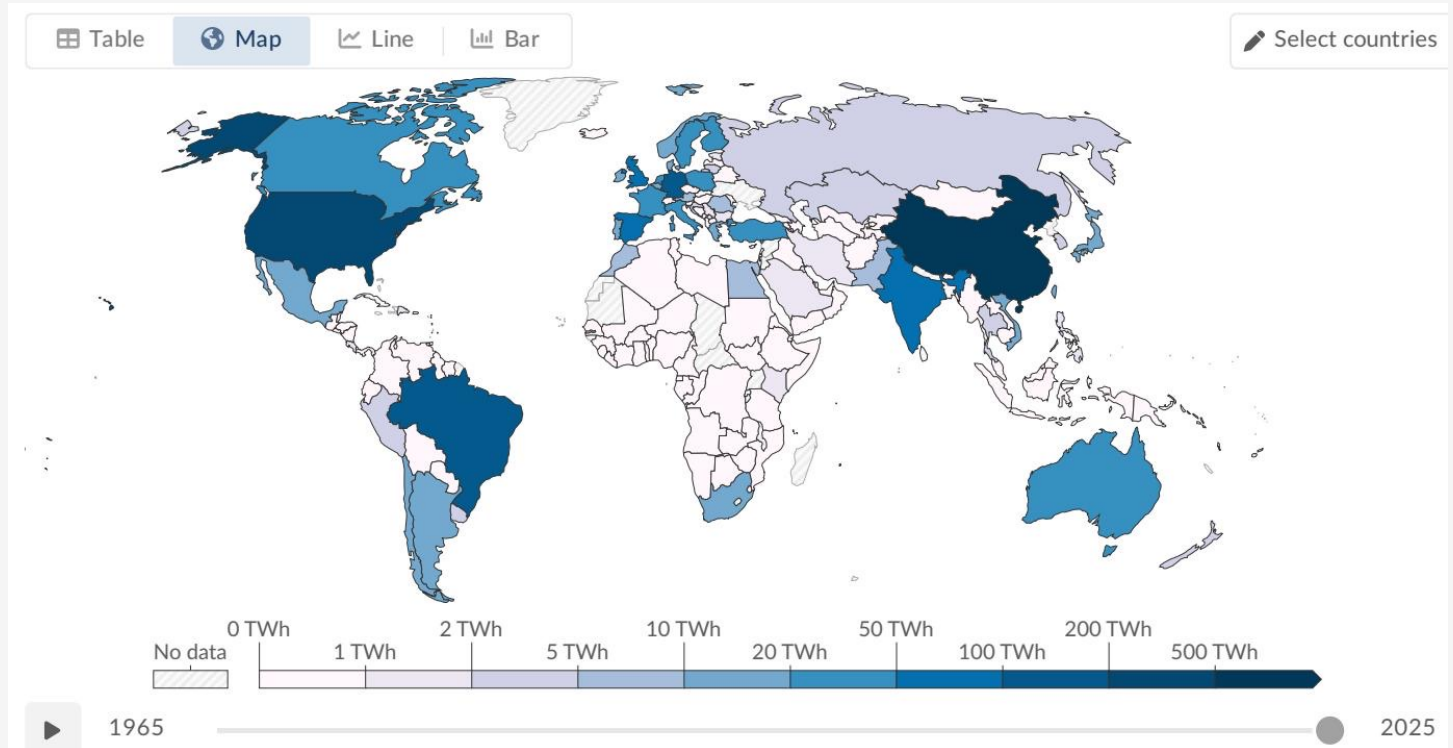


Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

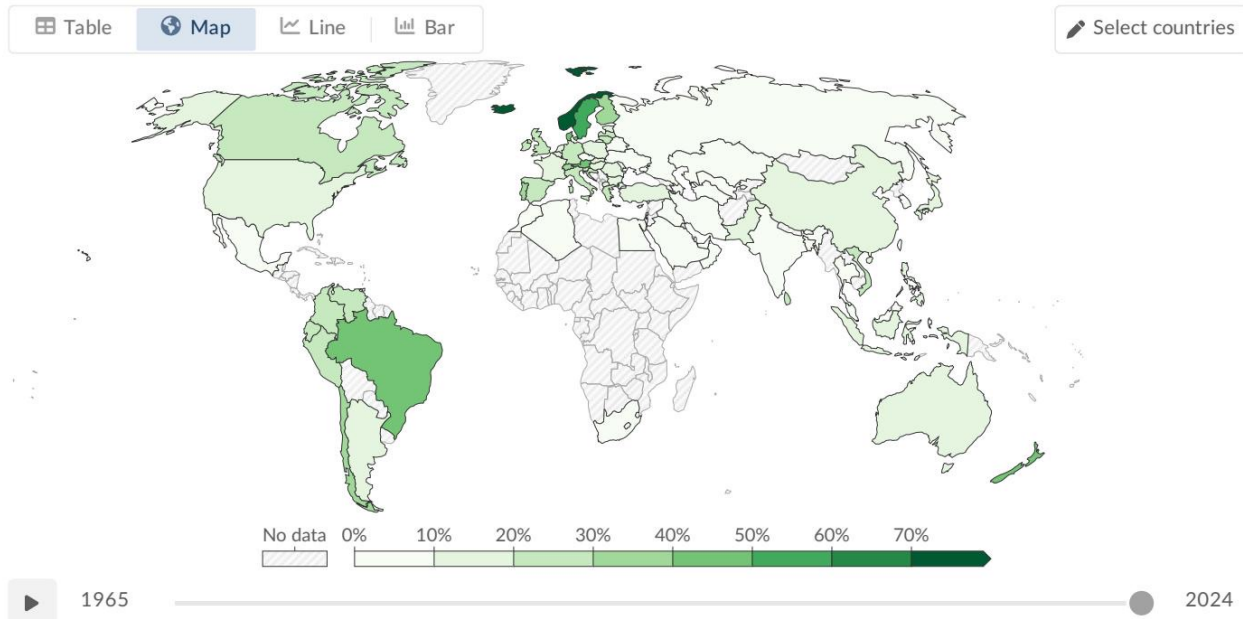
Several African and Middle Eastern countries have shares below 10%, reflecting fossil fuel dependence and limited renewable investment.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

- Hydropower accounts for roughly half of all renewable electricity generation (~4,500 TWh).
- It has been generating electricity at scale for over a century.
- Top producers include China, Brazil, and Canada.
- While hydro still leads, its growth potential is limited compared to wind and solar.



Wind and Solar Are the Fastest-Growing Energy Technologies in History

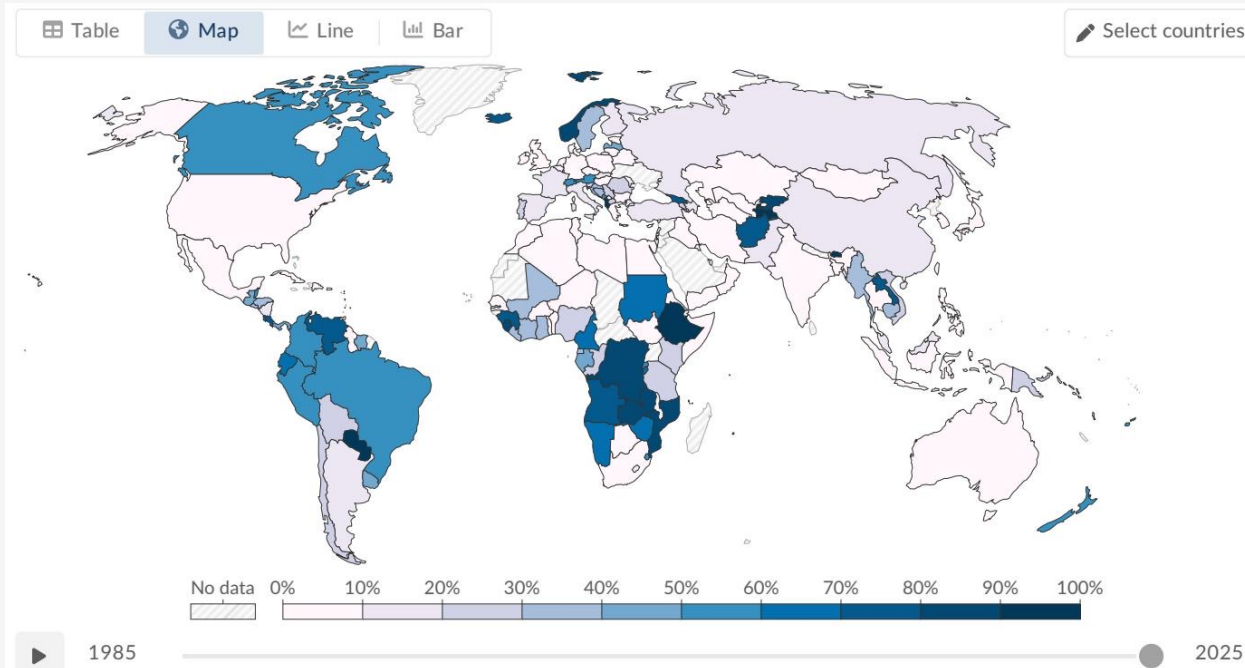


Wind vs. Solar: Generation & Share (2024)

Technology	Generation	Share of Renewables
Wind	~2,400 TWh	~25%
Solar	~2,000 TWh	~21%

- Solar is the fastest-growing source, with dramatic growth since 2010 driven by falling costs and policy support.
- Wind generation includes both onshore and offshore installations, especially strong in Europe, China, and the United States.
- Combined, wind and solar (~4,400 TWh) now nearly match hydropower's output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



- Scandinavian countries, Brazil, and Canada benefit from abundant hydropower (65–98% renewable electricity).
- European nations like Denmark and Germany have invested heavily in wind and solar.
- Much of Sub-Saharan Africa has minimal renewable electricity infrastructure.
- Many Middle Eastern nations remain almost entirely dependent on fossil fuels for electricity (<10% renewables).
- The transition is highly uneven — geography, policy, and investment drive the divide.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

- ~14% of global primary energy and ~30% of global electricity now come from renewables — a significant milestone, but far from sufficient for climate targets.
- Hydropower remains dominant (~47% of renewable electricity), but its growth potential is limited compared to wind and solar.
- Wind and solar are the fastest-growing energy technologies in history; combined, they now produce nearly as much electricity as hydropower.
- Stark regional disparities persist: Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%.
- Transport and heating lag far behind electricity in the renewable transition, keeping the overall primary energy share much lower than the electricity share.
- Accelerating deployment across all sectors and regions is essential to meet climate and air quality goals.